MATH 152 **QUIZ 9** NAME _____

Mrs. Bonny Tighe 25 points

11.9 TA_____Wed 11/16/2022

1. Find the power series representation for the following functions.

a) $f(x) = \dfrac{x^2}{1+4x^2}$ Find the interval of convergence for this series only.

b) $f(x) = \ln(x+3)$

c) $f(x) = \dfrac{1}{(2x+1)^2}$

2. a) Approximate the integral using power series and 4 non-zero terms.

$$\int_0^{0.2} \frac{2x^2}{4 - 2x^2}\, dx$$

3. Express the function as the sum of power series by first using partial fractions. Find the interval and radius of convergence. $f(x) = \dfrac{5 + x}{3 - 10x + 3x^2}$

4. Find the derivative of the given power series

$$f(x) = \sum_{k=0}^{\infty} \frac{(-1)^k (1 \cdot 3 \cdot 5 \cdots (2k+1))\ x^{2k+1}}{3^{k+1}\ k!}$$

$f'(x) =$

$$f(x) = \sum_{k=0}^{\infty} \frac{(-1)^k (1 \cdot 3 \cdot 5 \cdots (2k+1))\ x^{2k}}{7^{k}\ k!}$$

$f'(x) =$

5. Find the integral of the power series $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n+1} (x+2)^{n+3}}{2^n}$

6. Rewrite the series $f(x) = \displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n\ x^{2n+1}}{3^{n+1}\ n!}$

here $\displaystyle\sum_{n=0}^{\infty}$ _____

MATH 152 **QUIZ 9A** NAME _____

Mrs. Bonny Tighe 25 points

11.9 TA_____Wed 11/16/2022

1. Find the power series representation for the following functions.

a) $f(x) = \dfrac{x}{1+3x^3}$ Find the interval of convergence for this series only .

b) $f(x) = \ln(x+4)$

c) $f(x) = \dfrac{1}{(x-1)^2}$

2. Find the integral of the power series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}(x-2)^{n+2}}{5^{n+1}(2n)}$

3. Find the derivative of the given power series

$$f(x) = \sum_{k=0}^{\infty} \frac{(-1)^k (1 \cdot 3 \cdot 5 \cdots (2k+1)) \; x^{2k+1}}{3^{k+1} k!}$$

$f'(x) =$

$$f(x) = \sum_{k=0}^{\infty} \frac{(-1)^k (1 \cdot 3 \cdot 5 \cdots (2k+1)) \; x^{2k}}{7^k k!}$$

$f'(x) =$

4. Express the function as the sum of power series by first using partial fractions. Find the interval and radius of convergence. $f(x) = \frac{1+x}{2-5x+2x^2}$

5. Rewrite the series $f(x) = \sum_{n=1}^{\infty} \frac{(-1)^n \quad x^{2n+1}}{3^{n+1} \; n!}$ here

$\sum_{n=0}^{\infty}$ _____

6. Approximate the integral using power series and 4 non-zero terms.

$$\int_{0}^{0.2} \frac{x}{4+x^2}\,dx$$